

# Nhat-Phat Nguyen

Final-Year AI / Computer Vision Researcher

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Vietnam

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## Research Interests

3D Reconstruction & Neural Rendering (3D Gaussian Splatting) · Medical Image Analysis  
· Deep Learning · Object Detection & Edge AI

## Education

2023–Present **B.Eng. in Information Technology**, HCMC University of Technology and Engineering (HCM-UTE), Ho Chi Minh City, Vietnam, GPA: 3.67 / 4.0

Concentration in Artificial Intelligence and Computer Vision. Recipient of the HCM-UTE Talented Student Scholarship (2024) and Academic Excellence Scholarship (2024).

## Publications

### Conferences

**Nhat-Phat Nguyen**, Quoc-Hung Pham, Van-Dung Hoang<sup>†</sup>.

“Adaptive Density-Based Geometry Compression for Efficient 3D Gaussian Splatting.”  
The 8th International Conference on Green Technology and Sustainable Development (GTSD 2026), Springer Track. **(Accepted)**

Introduced a geometry-aware compression method using adaptive density estimation to remove spatially redundant Gaussian primitives while preserving critical structures, achieving a 30% Gaussian reduction while largely preserving average rendering quality across 11 scenes on two standard benchmarks.

### Preprints

**Nhat-Phat Nguyen**, Quoc-Hung Pham, Van-Dung Hoang<sup>†</sup>.

“Self-Supervised CT Through-Plane Interpolation via Residual 3D Gaussian Splatting.”  
Independent Research, 2026. **(Preprint)**

A self-supervised approach reconstructing high-resolution volumetric CT stacks from sparse axial slices without paired supervision, using residual 3D Gaussian Splatting.

<sup>†</sup> Corresponding author / research supervisor. **Bold** indicates self.

## Research & Selected Projects

2026 **Real-Time Construction Site Safety Monitoring System**, HCM-UTE – Faculty of Advanced Education, AI/ML Lead

Designed a zone-aware PPE compliance detection system on edge hardware (Jetson Xavier NX). YOLO26n-based model achieving 0.677 mAP@0.5 at 131 FPS with 99.93% bandwidth reduction versus cloud streaming. Awarded 3rd Prize, Edge Computing Innovation 2026.

- 2024–2025 **Advanced Retinal Vessel Analysis Using Deep Learning**, *HCM-UTE University Research Program, Lead Researcher*  
Conducted a deep-learning research project segmenting retinal blood vessels from medical images. Applied CNNs and image processing for high-resolution pixel-level medical image analysis. Rated “Good”; presented at Euréka XXVII 2025 (VNU-HCM).
- 2024 **Cassava Leaf Disease Classification**, *Independent Project*  
Developed a MobileNetV3 classifier with k-fold cross-validation and data augmentation, achieving over 90% validation accuracy, with a desktop GUI interface.

**Awards & Honors**

- Jul 2026 **1st Prize** – Project Showcase 2026, Academic Pathways Track (HCM-UTE)
- Jul 2026 **3rd Prize** – Edge Computing Innovation 2026 (Faculty of Advanced Education, HCM-UTE)
- Apr 2026 Outstanding Student in Scientific Research – FAEPRIME, HCM-UTE (AY 2024–2025)
- Jan 2026 Student of Five Merits – **Ho Chi Minh City Level**
- Mar 2026 Student of Five Merits – University Level (HCM-UTE)
- Oct 2025 Euréka XXVII 2025 – Student Scientific Research Prize (Vietnam National University HCM)
- Dec 2024 HCM-UTE Talented Student Scholarship
- Dec 2024 Academic Excellence Scholarship – Good Standing (Conduct score: 100/100)
- Jan 2024 Young Leader Face of Ho Chi Minh City 2023 (HCMC Youth Union)

**International Academic Exchange**

- Jan 2026 **BMW Program – Digital Twin & AI-Based Industrial Simulation**, *University of Ulsan, South Korea*  
Inbound academic exchange on digital twin technology and AI-based industrial simulation; collaborated with international students.

**Technical Skills**

|               |                                     |                    |                                                                     |
|---------------|-------------------------------------|--------------------|---------------------------------------------------------------------|
| Languages     | Python, C++, C#, SQL                | ML/DL Frameworks   | PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV |
| Tools & MLOps | GitHub, Docker, CUDA, Kaggle, Colab | Languages (spoken) | Vietnamese (Fluent), English (IELTS 6.0)                            |

**Leadership & Service**

- 2023–Present Youth Union Secretary, Class 23110FIE1, HCM-UTE – coordinated student activities and community engagement
- 2024–2025 Volunteer: Spring Volunteer Campaigns, Exam Season Support; volunteer social work score 99/100